

Homological Algebra - Self Study

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Abstract

I am currently reading Weibel's *Introduction to Homological Algebra* to better understand homological algebra. Since I already have an understanding of some of the fundamental concepts of homological algebra, I started with chapter 7 during spring 2026 due to two of my classes working through Lie algebra cohomology. I intend to read through the book starting in chapter 1 this summer, since homology is becoming increasingly relevant in my coursework and it would be beneficial to improve my understanding of the subject. [Wei94]

Chapter 1

Lie algebra homology and cohomology

1.1 Lie algebras

Let R be a commutative ring.

Definition 1.1.1. A *Lie R -algebra*, usually called a *Lie algebra*, is an R -module $\mathfrak{g}$ together with a *Lie bracket* $[-, -] : \mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$ often referred to as a multiplication operation, which is not necessarily associative, but instead satisfies the following:

- **(Antisymmetry)** $[x, x] = 0$;
- **(Jacobi identity)** $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

A morphism $\varphi : \mathfrak{g} \rightarrow \mathfrak{h}$ of Lie R -algebras is an R -linear map which preserves the bracket in the sense that the diagram

$$\begin{array}{ccc} \mathfrak{g} \times \mathfrak{g} & \xrightarrow{[-, -]_{\mathfrak{g}}} & \mathfrak{g} \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \\ \mathfrak{h} \times \mathfrak{h} & \xrightarrow{[-, -]_{\mathfrak{h}}} & \mathfrak{h} \end{array}$$

commutes. One can equivalently work with maps from the tensor products, of course.

Intuitively, the bracket is a commutator, and indeed every associative R -algebra A can be viewed as a Lie R -algebra by defining $[x, y] = xy - yx$. We write $\text{Lie}(A)$ for this Lie algebra, and thus we obtain a functor $\text{Lie} : \text{Alg}_R \rightarrow \text{Lie}_R$. This functor has a left adjoint given by the universal enveloping algebra $U : \text{Lie}_R \rightarrow \text{Alg}_R$, which we will explore indepth later.

An *ideal* is a R -submodule $\mathfrak{h} \subseteq \mathfrak{g}$ such that $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$. An ideal is immediately a Lie subalgebra, and one can quotient by an ideal to obtain a Lie algebra. Every ideal can thus be written as a short exact sequence of Lie algebras

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathfrak{h} \longrightarrow 0.$$

An *abelian* Lie algebra is a Lie algebra where $[-, -] = 0$.

Example 1.1.2. Given an associative R -algebra A , we obtain the Lie algebra $\mathfrak{gl}_n(A)$ of matrices with values in A , together with important Lie subalgebras

- $\mathfrak{sl}_n(A)$: the set of traceless matrices. This becomes difficult to define when A is noncommutative.
- $\mathfrak{o}_n(A)$: the set of skew-symmetric matrices $A_{ij} = -A_{ji}$.
- $\mathfrak{t}_n(A)$: the set of upper triangular matrices.

- $\mathfrak{n}_n(A)$: the set of strictly upper triangular matrices.

Example 1.1.3. Let A a not necessarily associative R -algebra. A derivation of A into itself is a R -linear map $D : A \rightarrow A$ satisfying a Leibnitz rule

$$D(ab) = (D(a))b + a(D(b)).$$

The set $\text{Der}(A)$ of derivations on A has a commutator $[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$, which makes $\text{Der}(A)$ into a Lie algebra.

Construction 1.1.4. Given an R -module M , the *free Lie algebra* $\mathfrak{f}(M)$ is a Lie algebra containing M as a submodule which is universal with respect to turning R -linear maps into Lie algebra maps. The free Lie algebra assembles into a functor $\mathfrak{f} : \text{Mod}_R \rightarrow \text{Lie}$ with $\mathfrak{f} \dashv U$ for U the forgetful functor. The existence of such a Lie algebra follows by the adjoint functor theorem.

Construction 1.1.5. Given Lie algebras $\mathfrak{g}, \mathfrak{h}$, the product Lie algebra $\mathfrak{g} \times \mathfrak{h}$ is defined by

$$[(g, h), (g', h')] = ([g, g'], [h, h']).$$

The *lower central sequence* of a Lie algebra is the descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^3 = [\mathfrak{g}^2, \mathfrak{g}] \supseteq \cdots \supseteq \mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}] \supseteq \cdots.$$

A Lie algebra is *nilpotent* if $\mathfrak{g}^n = 0$ for some n . For example, $\mathfrak{n}_n(A)$ is nilpotent, and so are abelian Lie algebras.

The *derived series* of a Lie algebra $\mathfrak{g}$ is the following descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^{(2)} = [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}] \supseteq \cdots \supseteq \mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}] \supseteq \cdots.$$

We say that $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n . For example, every nilpotent Lie algebra is solvable: simply show $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ and the statement follows inductively. The converse is not true – for example, $\mathfrak{t}_n(A)$ is solvable but not nilpotent.

1.2 $\mathfrak{g}$ -Modules

Definition 1.2.1. A (left) $\mathfrak{g}$ -module M is an R -module together with an R -bilinear multiplication map $\mathfrak{g} \otimes_R M \rightarrow M$ (written $x \otimes m \mapsto xm$) such that

$$[x, y]m = x(y m) - y(x m).$$

A $\mathfrak{g}$ -module homomorphism is an R -linear map $\varphi : M \rightarrow N$ such that $\varphi(xm) = x\varphi(m)$. We write $\text{Hom}_{\mathfrak{g}}(M, N)$ or $\text{Mod}_{\mathfrak{g}}(M, N)$ for the set of $\mathfrak{g}$ -module homomorphisms, yielding an abelian category $\text{Mod}_{\mathfrak{g}}$.

Remark 1.2.2. Note that $\text{Hom}_{\mathfrak{g}}(M, N) \subseteq \text{Hom}_R(M, N)$ is an R -submodule.

Example 1.2.3. Let A be an associative R -algebra. Then any A -module M is also a $\text{Lie}(A)$ -module by taking the action $\text{Lie}(A) \otimes_R M \rightarrow M$ to be the same as the action $A \otimes_A M \rightarrow M$.

Example 1.2.4. There is a trivial $\mathfrak{g}$ -module functor from $\text{Mod}_R \rightarrow \text{Mod}_{\mathfrak{g}}$ given by taking an R -module M , and defining the action by $\mathfrak{g}$ to be the trivial action $0 : \mathfrak{g} \otimes_R M \rightarrow M$. It is an exact functor, and elements of its image are called *trivial $\mathfrak{g}$ -modules*. Write this functor as $\mathbf{0}$. It has left and right adjoints.

The *invariant submodule* $M^{\mathfrak{g}}$ of a $\mathfrak{g}$ -module M is the submodule

$$M^{\mathfrak{g}} = \{m \in M \mid xm = 0 \forall x \in \mathfrak{g}\}.$$

We note that $M^{\mathfrak{g}} \cong \text{Mod}_{\mathfrak{g}}(R, M)$, when viewing R as a trivial R -module. This follows since R -linear maps $\varphi : R \rightarrow M$ are determined by $\varphi(1)$ since $\varphi(r) = r\varphi(1)$. To make it $\mathfrak{g}$ -linear, we need $x\varphi(1) = \varphi(x1) = \varphi(0) = 0$ since R is a trivial module, hence we may only map 1 to an element of $M^{\mathfrak{g}}$. Moreover such a choice uniquely determines φ . Since it is the maximal trivial submodule of $M^{\mathfrak{g}}$, we obtain $\mathbf{0} \dashv -^{\mathfrak{g}}$. This is immediate to see: a $\mathfrak{g}$ -module homomorphism $\mathbf{0}(N) \rightarrow M$ must identify $x\varphi(n) = 0$, so its image must fall into a trivial submodule, hence it factors through $M^{\mathfrak{g}}$. Moreover, any R -linear map $N \rightarrow M^{\mathfrak{g}}$ induces a map $\mathbf{0}(N) \rightarrow M$ in this way. Hence we have $\mathbf{0} \dashv -^{\mathfrak{g}}$. Additionally, since $M^{\mathfrak{g}} \cong \text{Hom}_{\mathfrak{g}}(\mathbf{0}(R), M)$, we have a representation $-^{\mathfrak{g}} \cong \text{Hom}_{\mathfrak{g}}(\mathbf{0}(R), -)$.

The *coinvariants* of M are the quotient $M/\mathfrak{g}M$. Note that this is also a trivial module since for any $x \in \mathfrak{g}$ and $m \in M$, we have $xm \in \mathfrak{g}M$ which is identified to 0 in the quotient. Since this is the largest quotient which is a trivial module, we can show that $-\mathfrak{g} \dashv \mathbf{0}$. We will see later that $\text{Mod}_{\mathfrak{g}}$ has enough projectives and injectives, hence we can compute the derived functors of (co)invariants.

Definition 1.2.5. Let M be a $\mathfrak{g}$ -module. We write $H_{\bullet}(\mathfrak{g}, M)$ or $H_{\bullet}^{\text{Lie}}(\mathfrak{g}, M)$ for the derived functors $L_{\bullet}(-\mathfrak{g})(M)$ of $-\mathfrak{g}$, and call them the *homology groups of $\mathfrak{g}$ with coefficients in M* . By definition, $H_0(\mathfrak{g}, M) = M_{\mathfrak{g}}$.

Similarly, write $H^{\bullet}(\mathfrak{g}, M)$ or $H_{\text{Lie}}^{\bullet}(\mathfrak{g}, M)$ for the right derived functors $R^{\bullet}(-^{\mathfrak{g}})(M)$ of $-^{\mathfrak{g}}$, and call them the *cohomology groups of $\mathfrak{g}$ with coefficients in M* . By definition, $H^0(\mathfrak{g}, M) = M^{\mathfrak{g}}$.

1.3 Universal Enveloping Algebras

Write $T(\mathfrak{g})$ for the tensor algebra on $\mathfrak{g}$. If $\mathfrak{g}$ is a Lie algebra, the universal enveloping algebra can be given as $T(\mathfrak{g})$ modulo the relation that the inclusion $\mathfrak{g} \hookrightarrow T(\mathfrak{g})$ be a Lie algebra homomorphism and $T(\mathfrak{g})$ have bracket given by the commutator. It is universal in the sense that precomposition by the inclusion yields

$$\text{Lie}_R(-, \text{Lie}) \cong \text{Alg}_R(U, -).$$

Proposition 1.3.1. *If $\mathfrak{g}$ is a Lie algebra, then every $\mathfrak{g}$ -module is naturally a $U\mathfrak{g}$ -module, and vice versa. This yields natural isomorphisms $\text{Mod}_{\mathfrak{g}} \cong \text{Mod}_{U\mathfrak{g}}$.*

Proof. Transposing under the tensor-hom adjunction shows that a $\mathfrak{g}$ -module is a Lie algebra map $\mathfrak{g} \rightarrow \text{Lie}(\text{End}_R(M))$ for some R -module M . A $U\mathfrak{g}$ -module is an algebra map $U\mathfrak{g} \rightarrow \text{End}_R(M)$. The statement now follows by adjointness

$$\text{Lie}_R(\mathfrak{g}, \text{Lie}(\text{End}_R(M))) \cong \text{Alg}_R(U\mathfrak{g}, \text{End}_R(M)).$$

■

Since the categories of associative algebras have enough projectives and injectives, proposition 1.3.1 shows that so do the categories of $\mathfrak{g}$ -modules. The action is simply given by, for $x_1, \dots, x_n \in \mathfrak{g}$,

$$(x_1 \cdots x_n)m = x_1(\cdots(x_nm)\cdots).$$

We then impose bilinearity to cover all of $U\mathfrak{g}$.

Corollary 1.3.2. *Let M be a $\mathfrak{g}$ -module. Then*

$$H_\bullet(\mathfrak{g}, M) \cong \mathrm{Tor}_\bullet^{U\mathfrak{g}}(R, M),$$

$$H^\bullet(\mathfrak{g}, M) \cong \mathrm{Ext}_{U\mathfrak{g}}^\bullet(R, M).$$

Proof. Since we have already observed that $M^\mathfrak{g} \cong \mathrm{Hom}_\mathfrak{g}(R, -) \cong \mathrm{Hom}_{U\mathfrak{g}}(R, -)$, the second statement is immediate by functoriality of derived functors (in the homotopy sense) and homology. For the first, note that the transpose of $0 : \mathfrak{g} \rightarrow R$ under $U \dashv \mathrm{Lie}$ is the unique map $U\mathfrak{g} \rightarrow R$ mapping $\mathfrak{g} \mapsto 0$. Define the *augmentation ideal* $\mathfrak{J}$ to be the kernel of this map; evidently $\mathfrak{J}$ is the two-sided ideal generated by the image of $\mathfrak{g}$ in $U\mathfrak{g}$. This map is epi, hence the first isomorphism theorem yields $R \cong U\mathfrak{g}/\mathfrak{J} \cong (U\mathfrak{g})_\mathfrak{g}$. Then

$$R \otimes_{U\mathfrak{g}} M \cong (U\mathfrak{g}/\mathfrak{J}) \otimes_{U\mathfrak{g}} M \cong M/\mathfrak{J}M \cong M/\mathfrak{g}M \cong M_\mathfrak{g}.$$

■

Theorem 1.3.3 (Poincaré-Birkhoff-Witt). *Let $\mathfrak{g}$ be a free R -module. Then $U\mathfrak{g}$ is also a free R -module. If $\{e_\alpha\}_{\alpha \in A}$ is an ordered basis of $\mathfrak{g}$, then the collection of all elements of the form e_I , where I is an increasing sequence, form a basis for $U\mathfrak{g}$.*

We decline to prove the PBW.

Corollary 1.3.4. *The inclusion $\mathfrak{g} \hookrightarrow U\mathfrak{g}$ is injective, so we may identify $\mathfrak{g}$ with its image.*

Corollary 1.3.5. *If $\mathfrak{h} \subseteq \mathfrak{g}$ is a Lie subalgebra, and $R = k$ is a field, then $U\mathfrak{g}$ is a free $U\mathfrak{h}$ -module.*

Exercise 1.3.6. Let M, N be left $\mathfrak{g}$ -modules. Show that $\mathrm{Mod}_R(M, N)$ is a $\mathfrak{g}$ -module by $(xf)(m) = xf(m) - f(xm)$. Then show that $\mathrm{Mod}_\mathfrak{g}(M, N) \cong \mathrm{Mod}_R(M, N)^\mathfrak{g}$. Show that this induces isomorphisms $\mathrm{Ext}_{U\mathfrak{g}}^\bullet(M, N) \cong H_{\mathrm{Lie}}^\bullet(\mathfrak{g}, \mathrm{Mod}_R(M, N))$. In particular, the global dimension of $U\mathfrak{g}$ equals the Lie algebra cohomological dimension of $\mathfrak{g}$.

Proof. For the first, we need only show $[x, y]f = x(yf) - y(xf)$. Let $x, y \in \mathfrak{g}$ and $m \in M$. Then since M, N are $\mathfrak{g}$ -modules,

$$\begin{aligned} ([x, y]f)(m) &= [x, y]f(m) - f([x, y]m) \\ &= xyf(m) - yxf(m) - f(xym - yxm) \\ &= xyf(m) - yxf(m) - f(xym) + f(yxm). \end{aligned}$$

Note that $xf = x \circ f - f \circ x$. Thus we also have

$$x(yf) = x(y \circ f - f \circ y) = x \circ y \circ f - y \circ f \circ x - x \circ f \circ y + f \circ y \circ x.$$

Hence evaluating at m yields

$$(x(yf))(m) = xyf(m) - yf(xm) - xf(ym) + f(yxm).$$

Similarly,

$$(y(xf))(m) = yxf(m) - xf(ym) - yf(xm) + f(xym).$$

Thus

$$\begin{aligned} & (x(yf))(m) - (y(xf))(m) \\ &= xyf(m) - yf(xm) - xf(ym) + f(yxm) - yxf(m) + xf(ym) + yf(xm) - f(xym) \end{aligned}$$

$$= xyf(m) + f(yxm) - yxf(m) - f(xym).$$

They agree, hence $\text{Mod}_R(M, N)$ is a $\mathfrak{g}$ -module in the way claimed.

Now let f be a $\mathfrak{g}$ -invariant of $\text{Mod}_R(M, N)$. Then $0 = xf = x \circ f - f \circ x$ for all $x \in \mathfrak{g}$, and thus $x \circ f = f \circ x$. Thus f is $\mathfrak{g}$ -linear. Conversely if f is $\mathfrak{g}$ -linear, then $f \circ x = x \circ f$, and thus $xf = x \circ f - f \circ x = 0$, so f is a $\mathfrak{g}$ -invariant of $\text{Mod}_R(M, N)$. $\blacksquare$

1.4 H^1 and H_1

Since $\mathfrak{J}$ is the kernel of $U\mathfrak{g} \rightarrow R$, there is a short exact sequence of $\mathfrak{g}$ -modules

$$0 \longrightarrow \mathfrak{J} \longrightarrow U\mathfrak{g} \longrightarrow R \longrightarrow 0.$$

Since $H_\bullet(\mathfrak{g}, -) \cong \text{Tor}_\bullet^{U\mathfrak{g}}(R, -)$, we apply the *dimension shifting technique* to take $H_n(\mathfrak{g}, -) = H_{n-1}(\mathfrak{g}, -)$: consider the long exact sequence

$$\cdots \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \cdots.$$

Since $U\mathfrak{g}$ is free as a $U\mathfrak{g}$ -module, it is flat, and hence its Tor modules vanish. Thus the short exact sequence simply reads

$$\cdots \longrightarrow 0 \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow 0 \longrightarrow \cdots.$$

The only way for this to be exact is if the map $\text{Tor}_n^{U\mathfrak{g}}(R, M) \rightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M)$ is an isomorphism for $n \geq 2$. Additionally, reading this in degree $n = 1, 0$, we obtain the exact sequence

$$0 \longrightarrow H_1(\mathfrak{g}, M) \longrightarrow \mathfrak{J} \otimes_{U\mathfrak{g}} M \longrightarrow M \longrightarrow M_{\mathfrak{g}} \longrightarrow 0.$$

Here we have noted that $R \otimes_{U\mathfrak{g}} M \cong M_{\mathfrak{g}}$.

Exercise 1.4.1. Show that the inclusion $i : \mathfrak{g} \rightarrow U\mathfrak{g}$ maps $[\mathfrak{g}, \mathfrak{g}]$ to $\mathfrak{J}^2$. Conclude that it induces a map $i : \mathfrak{g}^{ab} \rightarrow \mathfrak{J}/\mathfrak{J}^2$, where $\mathfrak{g}^{ab} = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$.

Proof. We consider the map $0 : \mathfrak{g} \rightarrow R$. $\blacksquare$

Bibliography

- [Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 1994.